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Evaluation you can audit: Eclipse PanEval as Shared Infrastructure

Michael Berns, Head of AI, Eclipse Foundation

AI 开放计算与智能体技术生态论坛 · Open Compute for AI and AI Agent Ecosystem Forum

FlagOS筑基开放算力，Agent启幕智能涌现 · FlagOS Anchors Open Compute, Agent Sparks Emergent Intelligence



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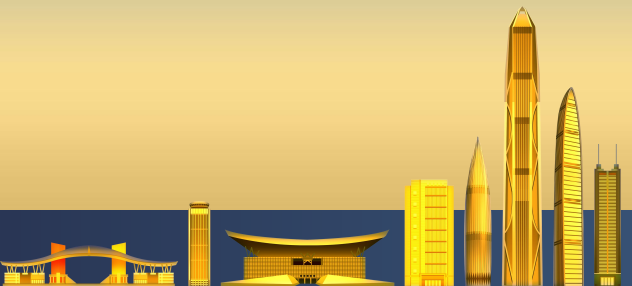
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01

Intro



A trusted home for global open source collaboration

Since 2004, competitors have built shared software here under vendor-neutral governance.

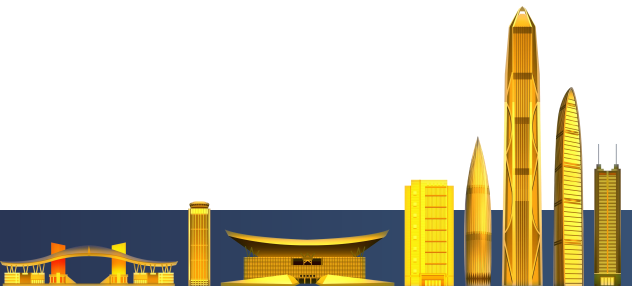
450+ PROJECTS

15K+ CONTRIBUTORS

300+ MEMBERS

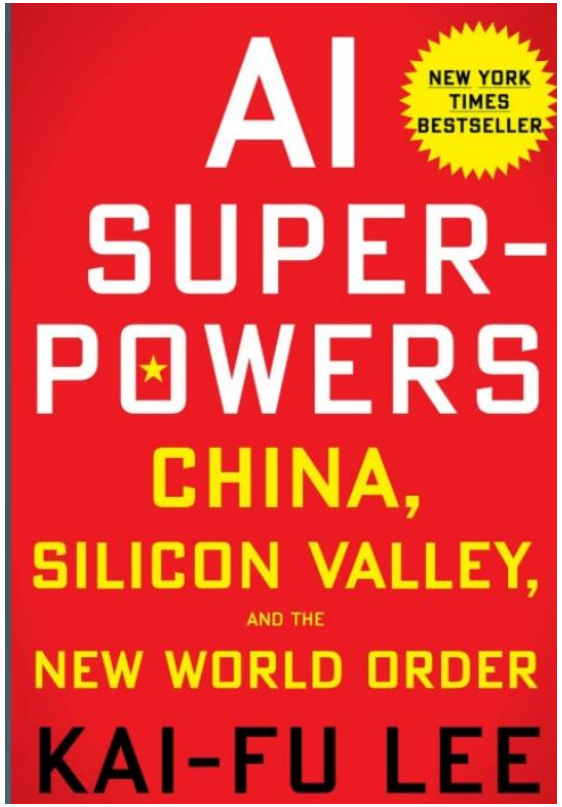
20+ COLLABORATIONS

Enterprise Java	Automotive & mobility	Embedded & IoT	Next-gen dev tooling	Other collaborations
ADOPTIUM	Eclipse SDV	OPENHW GROUP	eclipse	Open Regulatory Compliance
JAKARTA EE	openMDM	ECLIPSE THREADX		
MICROPROFILE		eclipse iot	ECLIPSE Cloud DevTools	AsciiDoc
OSGi	openMobility	oniro		
JAKARTA EE Future Directions	openPASS	Sparkplug	Open VSX	ECLIPSE DATASPACE




卧底经济学1
像经济学家那样思考
THE UNDERCOVER ECONOMIST

2011


AI
NEW YORK TIMES BESTSELLER
SUPER-POWERS
CHINA,
SILICON VALLEY,
AND THE
NEW WORLD ORDER
KAI-FU LEE

2018

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PanEval



Compliance Priorities and the Role of Eclipse PanEval

RISK-BASED FRAMEWORK

Unacceptable Risk	High Risk	Limited Risk	Minimal Risk
Prohibited AI Practices	Risk Management · Human Oversight · Data Governance	Transparency Obligations · AI Disclosure	Limited Obligations

WHERE ECLIPSE PANEVAL COULD PROVIDE IMPORTANT INSIGHTS RELEVANT TO EU AI ACT COMPLIANCE

■ Low ■■ Medium ■■■ High contribution

AI Inventory	Risk Classification	Safety Evaluation	Governance & Oversight	Documentation & Monitoring
LOW	MEDIUM	MEDIUM	LOW	HIGH

Key Dates

Apr 2021
First Draft Proposal

Aug 2024
Entered Into Force

Feb 2025
AI Literacy Requirements Begin

2 Aug 2026

Why It Matters

· Applies to providers and deployers serving the EU market
· Increased AI governance obligations
· Transparency & Accountability
· Growing influence on global AI governance

No open framework today does all three things at once

Aligned with EU conformity

Most benchmarks were designed for English-first, U.S.-centric research settings — not for the Article 95 obligations the EU AI Act now requires.

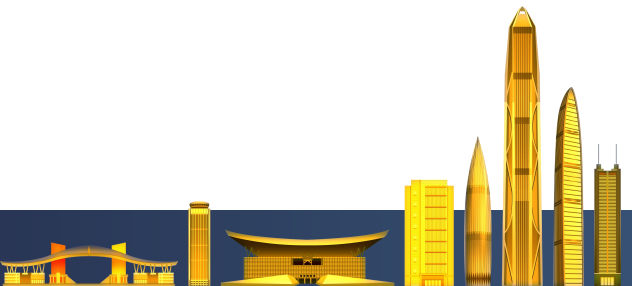
Covering safety and cyber together

Safety, bias, and robustness usually live in one place. Cybersecurity and vulnerability handling live in another. The AI Act treats them as one job.

Vendor-neutral, community-governed

Leaderboards and evaluation suites are often controlled by a single developer. Governance is the differentiator, not the metric.

Eclipse PanEval is designed to close this gap — open source, vendor neutral, and built for the EU regime.



Capacity × Task × Metrics — mapped to the obligations that matter

THE THREE-DIMENSIONAL MODEL

Capacity

Scope of model capabilities — what a model is asked to do.

Task

Benchmarks, datasets, adversarial tests that probe each capability.

Metrics

Accuracy, robustness, safety, fairness — multi-perspective scoring.

MAPPED TO THE AI ACT

Article 15 · Accuracy, robustness, cybersecurity

42 evaluation tasks across four modalities; Video-SafetyBench attack-success scoring; ROME and TRUE benchmarks for robustness.

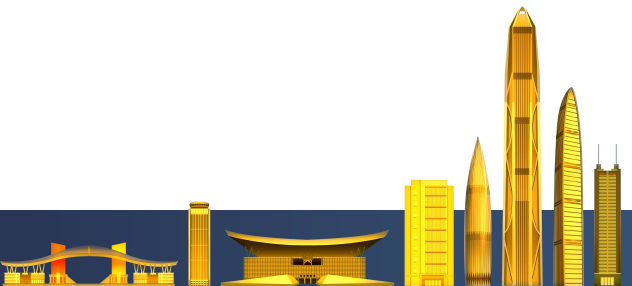
Article 53 · GPAI transparency

The Capacity–Task–Metrics framework maps directly to Annex XI evaluation evidence; structured JSON records.

Article 55 · Systemic risk

paneval-serving API for AI Office workflows; versioned task configs for a full audit trail.

PanEval supplies the evaluation evidence the AI Act asks for — it does not generate compliance documents itself.



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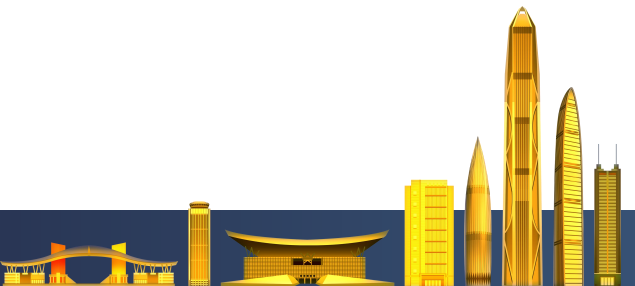
Other AI Projects



Eclipse Foundation AI Stack

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Applications & Use Cases	Vendor Independence & Sovereignty	Enterprise & Production Scale	Regulatory Readiness	Agent Ecosystem Development
Standards, Protocols & Registries	**EF AI Registry** Preview initiative, launched late 2025	**Jakarta Agentic AI** Spec ★29 · 12 fk · 201 commits · Nov '25		
Developer Tooling & Experience	**Eclipse Theia AI** ★21.5k · 2,800 forks · v1.72 May '26	**Eclipse Open VSX** ★1.9k · 300M+ dl/mo · 12k+ extensions	**Langium AI** ★977 · v4.2	**Prompt & Context Tooling** + Bring your project	**MCP / Skills Dev Kit** + Bring your project
Agent Runtime & Orchestration	**Eclipse LMOS** Production ★144 · Joined EF Dec '24	**Eclipse Graphene** 315 commits · 6 contributors (12mo)	**Agent Compliance Framework** + Bring your project	**Cross-agent Interoperability** + Bring your project	
Safety, Isolation & Evaluation	**Eclipse Enclave** Proposal submitted: May 2026	**Eclipse PanEval**	**Adversarial Testing & Red-teaming** + Bring your project		
Model Development & Foundation	**Eclipse DL4J** ★14.2k · 3,800 forks · Java + Scala	**Eclipse Aidge** 4,517 commits	**Eclipse January** ★36 · v2.3	**Model Provenance & Cards** + Bring your project	**Model Serving, Fine-tuning & RAG** + Bring your project



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Sovereign AI

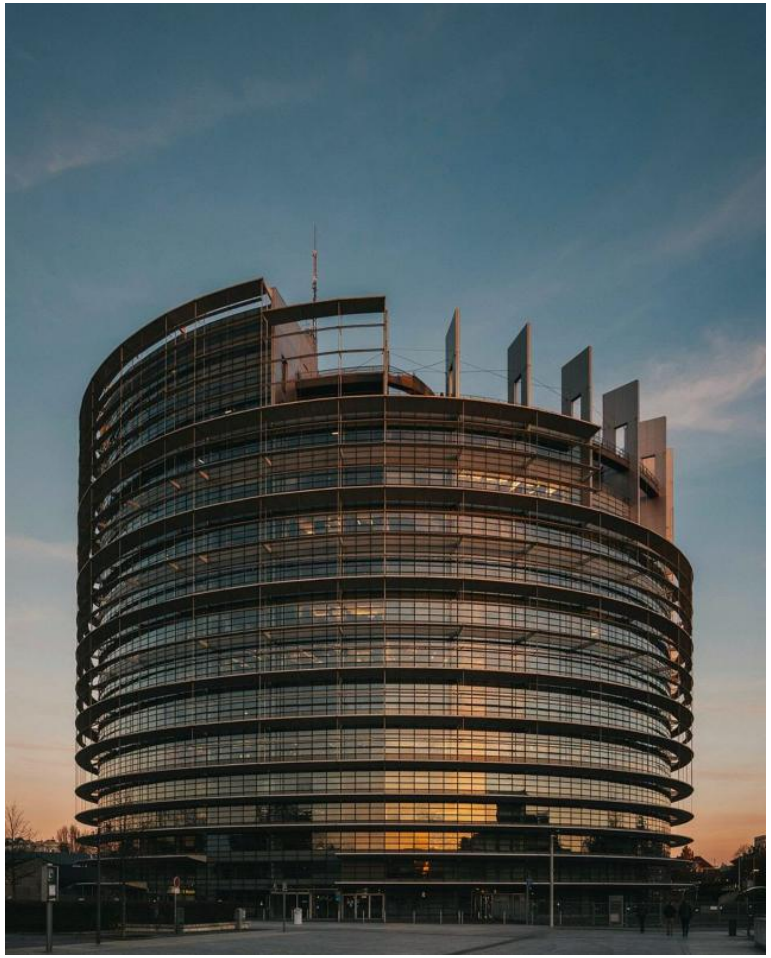


What the EC actually means by sovereignty

“We cannot afford to depend on others for the technologies that keep our hospitals running, our energy grids stable and our services secure. This is about protecting our citizens, defending our interests and making our own choices.”

“Together, we must turn these strengths into technological sovereignty.”

Ursula von der Leyen, June 2026 · Source: EC Press Release, European Technological Sovereignty Package, 3 June 2026



FOUR PILLARS

Semiconductor
s · AI · Cloud · Open source

THE REAL READ

Dependency and control, not geography of invention

NO CRISP DEFINITION

Sovereignty is used as a pillar, not a specification

Sovereignty across the AI stack

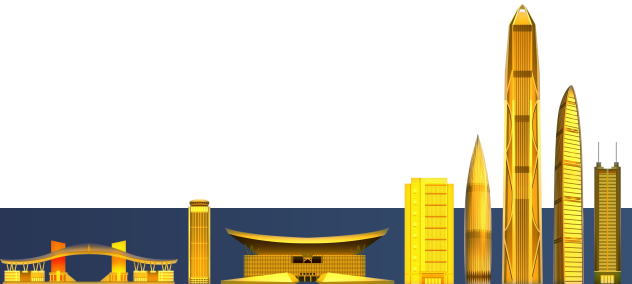
Theoretical freedoms vs practical freedoms: the distinction that matters.

	THEORETICAL — on paper	PRACTICAL — exercised today
Applications & skills	—	Build on open models; grow integration talent
Models & weights	Retrain from scratch — costly, slow, superseded within months	Fine-tune on your data; swap models freely
Tooling & MLOps	—	Independently evaluate and audit artefacts
Data & jurisdiction	Reproduce training runs — training data rarely released	Keep sensitive data under EU legal control
Compute & energy	—	Run where you choose — on-prem, private or EU cloud
Semiconductors	Build sovereign fabs — decade-scale, no near-term fix	No near-term practical lever

Governance — AI Act, CRA, procurement: cross-cutting

"[The EU is] not planning to work in isolation and produce everything ourselves." — Henna Virkkunen, June 2026

"A full retreat into a Europe-first tech approach will leave the continent weaker." — Keegan McBride, Tony Blair Institute



AI dependency is becoming a strategic risk

AI increasingly powers critical operations and decisions, while essential capabilities are becoming concentrated among a small number of providers.

DATA

Where and how information is processed

MODELS

Systems that may be difficult to inspect or adapt

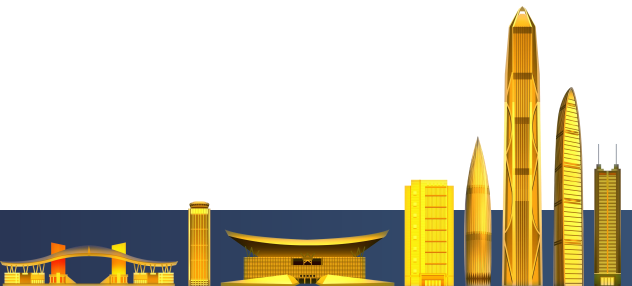
INFRASTRUCTURE

Workloads that may be difficult to move

TOOLS

Interfaces, skills, and supply chains controlled by individual providers

! **As AI becomes more deeply embedded, organisations risk losing the ability to change technologies, providers, and deployment choices.**



Sovereignty comes down to five practical freedoms

AI sovereignty means retaining the ability to understand, choose, adapt, and govern the AI systems on which an organisation depends.

DATA CONTROL

Control where and how data is processed

MODEL CHOICE

Choose models and infrastructure

PORTABILITY

Move workloads when needs change

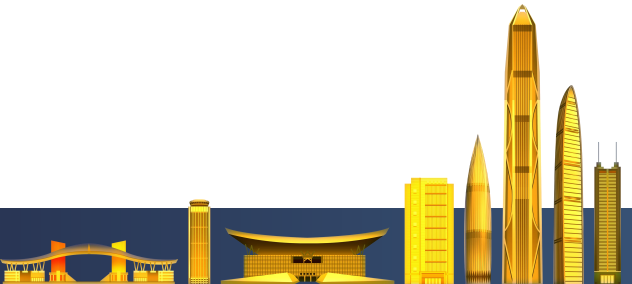
ADAPTABILITY

Adapt or replace components

OVERSIGHT

Maintain human oversight

Proprietary technology still has a place. Sovereignty means keeping the freedom to change it.



The Sovereign AI Foundation

Connecting member needs with open source AI projects and practical guidance

01 Map real-world needs and use cases

02 Evaluate relevant open source projects

03 Develop practical resources

04 Assess technologies and dependencies

05 Share experience and identify common requirements

06 Translate research into action

17 INITIAL PARTICIPATING ORGANISATIONS

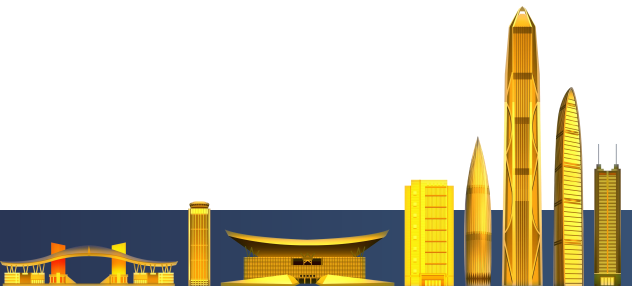
HOW THE ECOSYSTEM WORKS

Eclipse Foundation projects
Develop open source technologies and control their technical roadmaps

Sovereign AI Foundation
Connects member requirements, expertise, and feedback with relevant project communities

Eclipse Foundation
Provides the vendor-neutral governance and collaboration framework

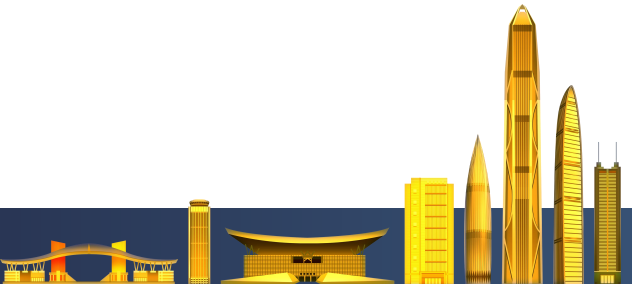
The Sovereign AI Foundation does not develop software or specifications or govern project roadmaps.



Launched 30 September 2026 • Open to all Eclipse Foundation member organisations

SAIF

SOVEREIGN AI FOUNDATION



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Thank you for inviting us

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